

Python Exercise - 2026/08/06

Question 01

- Write a Python program to input the first character of a shape:
 - R – Rectangle
 - T – Triangle
 - C – Circle
- The program should calculate and display area of the selected shape.
- Use the following inputs and formulas:

Shape	Inputs	Area Formula
Rectangle (R)	Width, Length	Area=Width × Length
Triangle(T)	Base, Height	Area=(Base × Height)/2
Circle (C)	Radius	Area= $\pi \times \text{Radius}^2$

- Requirements:
 1. Ask the user to enter first character of the shape(R,T or C)
 2. Based on the selected shape, input the required values.
 3. Calculate the area using the correct formula.
 4. Display the calculated area.

❖ Program

```
import math

shape = input("Enter the first character of the shape(R-Rectangle,T-Triangle,C-Circle):")
area = 0.0

if shape == "R" or shape == "r":
    length = float(input("Enter the length of rectangle:"))
    width = float(input("Enter the width of rectangle:"))
    area = length * width
elif shape == "T" or shape == "t":
    base = float(input("Enter the base of the triangle:"))
    height = float(input("Enter the height of the triangle:"))
    area = 0.5 * base * height
elif shape == "C" or shape == "c":
    radius = float(input("Enter the radius of the circle:"))
    area = math.pi * radius * radius
else:
    print("Invalid shape entered")

print("Shape area is", area)
```

❖ Outputs

✓ Correct Cases

```
Enter the first character of the shape(R-Rectangle,T-Triangle,C-Circle):R
Enter the length of rectangle:20
Enter the width of rectangle:6
Shape area is 120.0
```

```
Enter the first character of the shape(R-Rectangle,T-Triangle,C-Circle):T
Enter the base of the triangle:4
Enter the height of the triangle:12
Shape area is 24.0
```

```
Enter the first character of the shape(R-Rectangle,T-Triangle,C-Circle):C
Enter the radius of the circle:14
Shape area is 615.7521601035994
```

✓ Wrong Case

```
Enter the first character of the shape(R-Rectangle,T-Triangle,C-Circle):S
Invalid shape entered
Shape area is 0.0
```

Question 02

- An Adventure Resort offers the following activity packages:

Package Type	Title	Cost per Person (Rs.)
1	Horse Ride	2000
2	Scuba Diving	5000
3	Water Rafting	7000

- Write a Python program to:
 - Input the package type (H, S or W) from the keyboard.
 - Input the number of people.
 - Use if...elif...else statements to determine the cost per person.
 - Calculate the total cost using the formula:
 - Total cost = Cost per Person × Number of People
 - Display:
 - Package Title
 - Cost per Person
 - Number of People
 - Total Cost
- Note: Use if... elif... else statements only.

❖ Program

```
print("Adventure Resort")
print("-----")
print("\nAvailabe Package:")
print("1.Horse Ride")
print("2.Scuba Diving")
print("3.Water Rafting")

package=int(input("Enter the package type(1-3):"))
people=int(input("Enter the number of people:"))

if package==1:
    total=people*2000
elif package==2:
    total=people*5000
elif package==3:
    total=people*7000
else:
    total=0
    print("Invalid package")
if total>0:
    print("Total cost is:",total)
```

❖ Outputs

✓ Correct Cases

```
Adventure Resort
-----

Availabe Package:
1.Horse Ride
2.Scuba Diving
3.Water Rafting
Enter the package type(1-3):1
Enter the number of people:5
Total cost is: 10000
```

```
Adventure Resort
-----

Availabe Package:
1.Horse Ride
2.Scuba Diving
3.Water Rafting
Enter the package type(1-3):2
Enter the number of people:5
Total cost is: 25000
```

```
Adventure Resort
-----

Availabe Package:
1.Horse Ride
2.Scuba Diving
3.Water Rafting
Enter the package type(1-3):3
Enter the number of people:5
Total cost is: 35000
```

✓ Wrong Case

```
Adventure Resort
-----

Availabe Package:
1.Horse Ride
2.Scuba Diving
3.Water Rafting
Enter the package type(1-3):4
Enter the number of people:5
Invalid package
```

Question 03

- Write a python program to input values for three angles and display whether a triangle can be formed or not using given angle values.
- Hint: To form a triangle, the summation of all three angles should be equal to 180 and all three angles have to greater than 0 degrees.

❖ Program

```
angle1=int(input("Enter first angle:"))
angle2=int(input("Enter second angle:"))
angle3=int(input("Enter third angle:"))

if angle1+angle2+angle3==180 and angle1>0 and angle2>0 and angle3>0:
    print("Triangle can be formed")
else:
    print("Triangle cannot be formed")
```

❖ Outputs

✓ Correct Cases

```
Enter first angle:45
Enter second angle:80
Enter third angle:30
Triangle cannot be formed
```

✓ Wrong Case

```
Enter first angle:90
Enter second angle:80
Enter third angle:110
Triangle cannot be formed
```

Question 04

- Create a simple calculator that perform basic arithmetic operations based on the user's choice:

❖ Program

```
print("Simple Calculator")
print("-----")
num1=int(input("Enter first number:"))
num2=int(input("Enter second number:"))
print("\nChoose an Operation")
print("1.Addition")
print("2.Subtraction")
print("3.Multiplication")
print("4.Division")

operation=int(input("Enter the Operation(1-4):"))

if operation==1:
    Result=num1+num2
elif operation==2:
    Result=num1-num2
elif operation==3:
    Result=num1*num2
elif operation==4:
    Result=num1/num2
if operation>=1 and operation<=4:
    print("Result is:", Result)
else:
    print("Invalid Operation")
```

❖ Outputs

✓ Correct Case

```
Simple Calculator
-----
Enter first number:10
Enter second number:5

Choose an Operation
1.Addition
2.Subtraction
3.Multiplication
4.Division
Enter the Operation(1-4):3
Result is: 50
```

✓ Wrong Case

```
Simple Calculator
-----
Enter first number:20
Enter second number:5

Choose an Operation
1.Addition
2.Subtraction
3.Multiplication
4.Division
Enter the Operation(1-4):-1
Invalid Operation
```

Python Exercise – 2026/08/10

Question 01

- Write a python program to display all even numbers between 0 and 20.

❖ Program

```
for i in range(0,21):  
    if i % 2 == 0:  
        print(i)
```

```
for i in range(0,21,2):  
    print(i)
```

❖ Output

```
0  
2  
4  
6  
8  
10  
12  
14  
16  
18  
20
```

Question 02

- Write a python program to input the mark of 10 students and calculate the total and the average marks.

❖ Program

```
total=0
count=0

while count<10:
    mark=int(input("Enter mark:"))
    total=total+mark
    count=count+1
average=total/10

print("Total = ",total)
print("Average = ",average)
```

❖ Output

```
Enter mark:50
Enter mark:20
Enter mark:50
Enter mark:70
Enter mark:49
Enter mark:90
Enter mark:38
Enter mark:72
Enter mark:94
Enter mark:77
Total = 610
Average = 61.0
```

Question 03

- Write a program that prompts the user to enter a number and displays the multiplication table for that number up to a specified range.

❖ Program

```
number=int(input("Enter a number:"))
range_value=int(input("Enter the range:"))

multiplier=1

while multiplier<=range_value:
    result=number*multiplier
    print(number,"X",multiplier,"=",result)
    multiplier +=1
```

❖ Output

```
Enter a number:9
Enter the range:12
9 X 1 = 9
9 X 2 = 18
9 X 3 = 27
9 X 4 = 36
9 X 5 = 45
9 X 6 = 54
9 X 7 = 63
9 X 8 = 72
9 X 9 = 81
9 X 10 = 90
9 X 11 = 99
9 X 12 = 108
```

Question 04

- Write a Python program to input student's marks one by one until the user enters -1. Then calculate and display the total marks and the average marks. If no marks are entered, display "No grades were entered".

❖ Program

```
total=0
count=0

#Get first input from the user
grade=int(input("Enter mark,-1 to end:"))

while grade !=-1:
    total=total+grade
    count=count+1

    #Get next input
    grade=int(input("Enter mark,-1 to end:"))

if count !=0:
    average=total/count
    print("Class average is:",average)
else:
    print("No grades were entered.")
```

❖ Output

✓ Correct Case

```
Enter mark,-1 to end:50
Enter mark,-1 to end:39
Enter mark,-1 to end:79
Enter mark,-1 to end:89
Enter mark,-1 to end:98
Enter mark,-1 to end:-1
Class average is: 71.0
```

Wrong Case

```
Enter mark,-1 to end:-1
No grades were entered.
```

Question 05

- Write a Python program that prompts the user to enter a password. The user can attempt to enter the password a maximum of 3 times. If the correct password (password123) is entered, display "Access Granted". If all 3 attempts are used without entering the correct password, display "Access Denied". Use a sentinel-controlled repetition (or an equivalent loop in python).

❖ Program

```
correct_password="password123"
password=""
attempt=0

while password != correct_password and attempt<3:
    password = input("Enter password:")

    if password != correct_password:
        attempt=attempt+1
        print("Incorrect password.Please try a gain.")

if password == correct_password:
    print("Access granted.Welcome!")
else:
    print("Access denied.You have entered the password 3 times.")
```

❖ Output

- ✓ Correct Case

```
Enter password:password123
Access granted.Welcome!
```

- ✓ Wrong Case

```
Enter password:password
Incorrect password.Please try a gain.
Enter password:Password123
Incorrect password.Please try a gain.
Enter password:Passwod123
Incorrect password.Please try a gain.
Access denied.You have entered the password 3 times.
```

Question 06

- Write a program that prompts the user to enter a positive integer number. The program should calculate the sum of all numbers 1 to the entered number using a counter-controlled repetition with a do-while loop. Display the final sum.

❖ Program

```
number=int(input("Enter a positive integer number:"))
sum_of_numbers=0
count=1

while True:
    sum_of_numbers=sum_of_numbers+count
    count=count+1

    if count>number:
        break

print(f"The sum of numbers from 1 to {number} is: {sum_of_numbers}")
```

❖ Output

```
Enter a positive integer number:9
The sum of numbers from 1 to 9 is: 45
```

Question 07

- Write a program that prompts the user to enter a positive integer number, the program should calculate and display the factorial of the entered number.

❖ Program

```
number=int(input("Enter a positive integer number:"))

factorial=1

for i in range(1,number+1):
    factorial=factorial*i

print(f"The factorial of {number} is:{factorial}")
```

❖ Output

```
Enter a positive integer number:23
The factorial of 23 is:25852016738884976640000
```

Python Exercise – 2026/08/13

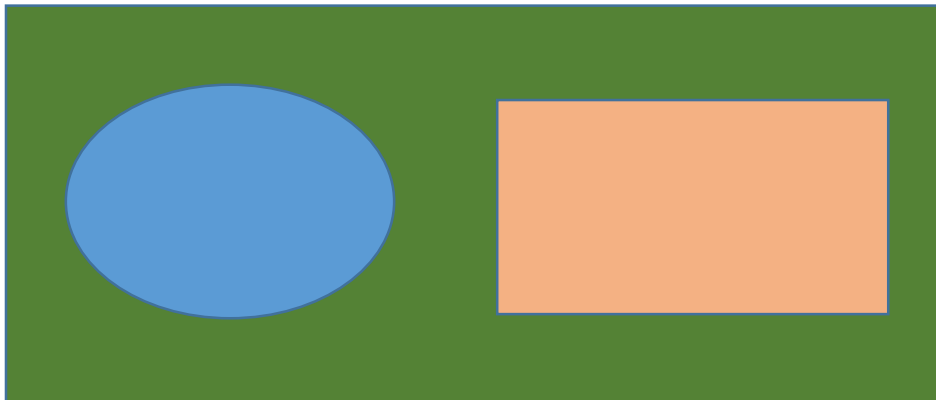
Question 01

- Write a Python program to calculate the green-colored area where grass needs to be planted. Use Area Of Green (), Area Of Rectangle (), Area Of Circle () function to do it. And assume owner need to build fence therefore find the perimeter of the yard.

- Hint:

Shape	Inputs	Area
Rectangle	Width, Length	Width × length
Circle	Radius	$\pi \times \text{radius} \times \text{radius}$

- Use appropriate functions, input values from the user, and display the calculated green and the perimeter of the yard.



❖ Program

```
def area_of_rectangle():
    width=float(input("Enter width:"))
    length=float(input("Enter length:"))
    area_r=width*length
    return area_r

def area_of_circle():
    radius=float(input("Enter radius:"))
    pi=22/7
    area_c=pi*radius*radius
    return area_c

def area_of_green(area_y,area_b,area_p):
    area_g=area_y - (area_b + area_p)
    print("Area need to plant grass is:",area_g)

def perimeter_of_yard():
    print("\n To find perimeter of yard enter width and length")
    width=float(input("Enter width:"))
    length=float(input("Enter length:"))
    perimeter=2*(width+length)
    print("perimeter of yard:",perimeter)

#Main function
print("Enter width and length of yard")
area_y=area_of_rectangle()

print("Enter width and length of building")
area_b=area_of_rectangle()

print("Enter radius of pond")
area_p=area_of_circle()

if(area_b+area_p)>area_y:
    print("Invalid value")
else:
    area_of_green(area_y,area_b,area_p)
```

❖ Output

✓Correct Case

```
Enter width and length of yard
Enter width:50
Enter length:50
Enter width and length of building
Enter width:4
Enter length:6
Enter radius of pond
Enter radius:7
Area need to plant grass is: 2322.0
```

✓Wrong Case

```
Enter width and length of yard
Enter width:10
Enter length:10
Enter width and length of building
Enter width:4
Enter length:6
Enter radius of pond
Enter radius:7
Invalid value
```